

ARJUN VARMA

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EDUCATION

Columbia University	New York, NY
Master of Science in Data Science, 3.6/4	Dec 2026
TA for Business Analytics II: Foundations of AI at Columbia Business School	
Vellore Institute of Technology	Vellore, IN
Bachelor of Technology in Electronics and Communication Engineering, 4.0/4.0	May 2022
Recipient of the Special Achiever Award	

WORK EXPERIENCE

ZS Associates	Pune, IN
Advanced Data Science Associate Consultant	Feb 2025 - Jun 2025
- Built and deployed an organization-wide analytics + ML platform consolidating multiple data sources to surface real-time KPIs by territory and product; partnered with PMs and marketing heads for a >10B REVENUE oncology portfolio - Reduced stakeholder prep time from days to minutes by replacing Excel-based reporting with automated dashboards - Piloted a retrieval-augmented LLM to turn FDA approval documents into concise briefs for the commercial teams	
ZS Associates	Pune, IN
Decision Analytics Associate Consultant	Jul 2024 - Jan 2025
- Led 5-member team to modernize legacy business rules; saved ~50 hrs/mo and improved first-pass quality to >99% - Built Positive-Unlabeled (PU) learning models to infer missing categorical labels in transactional medical data; expanded dataset coverage for customer journey analysis from ~40% to ~95% - Implemented automated model drift monitoring and unit testing for production ML pipelines - Secured a Top ~10% finish in company-wide hackathon; selected for lateral transfer to Data Science vertical	
ZS Associates	Pune, IN
Decision Analytics Associate	Feb 2022 - Jun 2024
- Engineered PySpark/SQL ETL pipelines integrating multiple data sources for \$4B+ oncology drug performance analytics - Defined patient-cohort inclusion/exclusion rules robust to missing/miscoded fields; delivered audit-ready analytics - Drove ad-hoc analytics surfacing care gaps and market opportunities for brand strategy across multiple launches - Promoted to Associate Consultant in 4 cycles (typical: 5); received Expert Associate and Insight Illuminator awards	

PROJECT EXPERIENCE

Financial RAG Chatbot - Python, FastAPI, ChromaDB, Streamlit	Github/Web Link
- Built LLM-powered RAG chatbot answering company financial queries from SEC filings with line-level citations - Implemented semantic retrieval using ChromaDB + text-embedding-3-large with automatic ticker/period parsing - Developed multi-model evaluation pipeline using Claude Opus as judge; deployed live demo on Streamlit Cloud	
Agricultural Product Standardization & Risk Detection - Python, GPT-4, RAG, REST API	
- Built RAG-augmented classification system with custom prompt engineering for Series-B East African agtech; categorized 7M+ transactions to assess farmer creditworthiness for microloans - Achieved 99% holdout accuracy via hybrid rule-based + GPT-5 (Transformer) RAG pipeline; reduced manual review load by 80%	
Biliary Tract Cancer (BTC) early detection using ML - Python, scikit-learn, PySpark	
- Developed predictive model for early Biliary Tract Cancer detection across 250M patient claims; enabled 45-day advance identification vs. standard diagnosis lag - Engineered hybrid feature pipeline: clinical risk factors, K-means/GMM patient segmentation, and Transformer-based NLP clustering on diagnosis narratives - Presented methodology at PMSA 2025 conference; model adopted for territory-level resource planning	
Seance AI - Historical Figure Chatbot - Python, Flask, OpenRouter API - sceance-ai.up.railway.app	Github/Web Link
- Built AI chatbot enabling conversations with 60+ historical figures using multi-model LLM support and streaming responses - Implemented era-appropriate prompt engineering and "Dinner Party" mode for multi-figure conversations; deployed on Railway	

SKILLS

Programming: Python, SQL, C++, R
Analytics & ML: Machine Learning, Deep Learning, Data Engineering, Extract Transform Load, Prompt engineering
Big Data & MLOps: PySpark/Spark, Databricks, MLflow, AWS (S3, EMR, Athena, SageMaker)
Libraries: pandas, NumPy, scikit-learn, matplotlib, BeautifulSoup, SHAP, PyTorch
Tools: Git, Jupyter, Claude Code, Cursor IDE, Jira/Confluence